

DONCASTER SHEFFIELD, United Kingdom

WMO: 034054

Lat: 53.475N

Lon: 1.004W

Elev: 17

StdP: 101.12

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.0	-5.7	2.3	-1.4	-4.2	2.7	0.1	14.7	9.0	13.4	9.1	2.3	170	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	27.1	18.7	25.1	18.0	23.1	17.0	19.6	24.9	18.6	23.4	17.7	22.1	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.8	21.8	16.9	12.1	20.9	16.0	11.4	19.6	56.0	24.7	52.8	23.5	50.2	22.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.5	9.3	DB	-6.3	30.8	3.7	2.0	-8.9	32.2	-11.1	33.4	-13.2	34.5	-15.9	35.9	(4)
(5)			WB	-6.5	21.6	3.6	1.4	-9.1	22.6	-11.2	23.4	-13.2	24.1	-15.8	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	4.8	5.0	6.8	9.3	12.2	15.2	17.6	16.9	14.5	11.4	7.5	5.4
	DBStd	5.41	3.05	3.35	3.04	2.91	2.94	2.57	2.57	2.25	2.47	2.95	3.35	3.69
	HDD10.0	724	162	141	108	47	12	0	0	0	1	18	89	146
	HDD18.3	2874	419	372	359	272	190	99	46	58	118	216	326	400
	CDD10.0	937	1	3	7	26	81	157	237	213	136	62	13	4
	CDD18.3	45	0	0	0	0	1	5	23	13	3	1	0	0
	CDH23.3	388	0	0	0	2	13	62	207	84	17	4	0	0
	CDH26.7	74	0	0	0	0	1	11	43	16	3	1	0	0
Wind	WSAvg	4.5	4.9	4.8	5.0	4.4	4.4	4.1	3.9	4.3	4.2	4.3	4.6	4.9
Precipitation	PrecAvg	575	48	35	42	45	47	51	49	53	54	46	52	50
	PrecMax	801	116	105	106	101	159	154	150	116	147	138	126	171
	PrecMin	387	9	1	7	6	9	5	10	4	10	8	20	8
	PrecStd	99	23	24	23	26	32	34	31	27	34	28	26	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.0	17.8	22.1	25.1	28.2	30.2	28.8	25.8	21.1	17.0	15.0
		MCWB	12.0	11.7	12.1	14.1	17.3	19.2	19.6	18.9	18.8	17.4	14.7	12.5
	2%	DB	12.1	12.9	15.1	18.9	22.3	25.1	27.8	25.6	22.8	18.9	15.2	13.8
		MCWB	10.1	10.2	11.2	12.8	16.0	17.6	19.1	18.4	17.4	16.1	13.3	11.8
	5%	DB	10.9	11.2	13.2	16.8	20.1	22.8	25.9	23.4	20.9	17.1	14.0	12.2
		MCWB	9.0	9.0	10.1	12.2	14.5	16.5	18.0	17.3	16.3	14.8	12.2	10.5
	10%	DB	9.8	10.1	12.0	15.0	18.1	21.0	23.7	21.9	19.1	16.0	12.8	11.0
		MCWB	8.3	8.2	8.9	11.1	13.2	15.8	17.4	16.4	15.1	13.7	11.0	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	11.7	12.9	15.5	18.3	20.7	21.2	20.2	19.5	17.6	14.9	12.9
		MCDB	14.1	13.8	15.7	20.3	23.7	26.5	26.9	25.3	23.8	20.9	16.7	14.4
	2%	WB	10.6	10.6	11.6	13.8	16.6	18.3	19.9	19.1	18.0	16.3	13.6	12.0
		MCDB	12.0	12.6	14.6	17.5	21.3	22.9	25.4	23.9	21.7	18.5	15.1	13.6
	5%	WB	9.4	9.6	10.6	12.5	15.2	17.1	18.9	18.1	16.9	15.2	12.5	11.0
		MCDB	10.6	11.3	13.2	16.1	19.3	21.4	24.0	22.5	19.8	16.9	13.9	12.6
	10%	WB	8.1	8.5	9.4	11.2	13.9	16.1	17.9	17.3	15.8	14.1	11.2	9.6
		MCDB	9.5	10.2	11.7	14.6	17.2	20.1	22.6	20.9	18.4	15.8	12.5	10.9
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.3	7.8	9.1	9.3	9.3	10.0	9.3	8.7	7.3	6.3	5.8
		MCDBR	6.6	7.1	10.0	11.9	12.9	13.2	13.6	11.8	10.7	8.0	7.0	6.5
		MCWBR	5.9	5.5	6.9	7.1	7.4	7.1	6.0	5.8	6.2	5.7	6.0	5.7
	5% WB	MCDBR	6.7	7.0	9.6	11.1	12.0	11.4	11.9	10.4	9.4	7.6	7.0	6.5
		MCWBR	6.1	5.7	7.0	7.2	7.2	6.9	6.1	5.8	6.2	5.7	6.1	5.8
Clear-Sky Solar Irradiance	taub	0.324	0.340	0.371	0.399	0.399	0.404	0.405	0.391	0.371	0.352	0.325	0.311	
	taud	2.347	2.385	2.296	2.232	2.271	2.283	2.288	2.347	2.410	2.442	2.402	2.323	
	Ebn at Noon	640	758	804	828	849	846	838	829	800	735	637	576	
	Edh at Noon	59	77	102	123	125	125	122	109	91	72	55	50	
All-Sky Solar Radiation	RadAvg	0.67	1.34	2.42	3.74	4.70	4.86	4.73	3.90	2.87	1.64	0.82	0.53	
	RadStd	0.06	0.15	0.29	0.45	0.31	0.46	0.49	0.26	0.26	0.14	0.09	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page